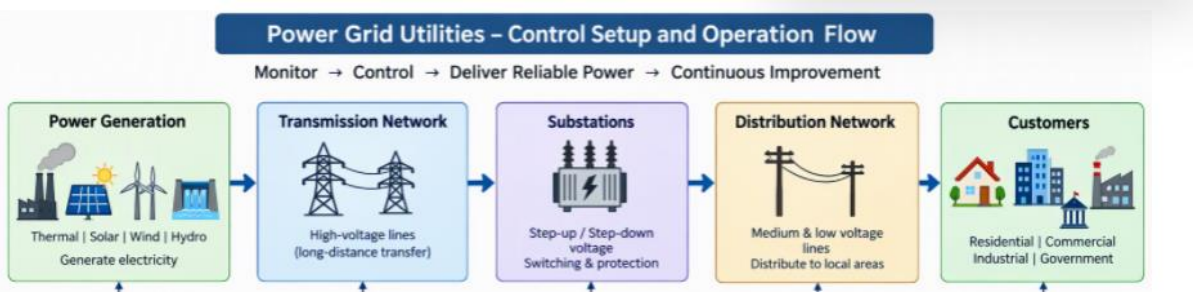
Capstone Project – Pre-Submission Document

IITM–JARO Batch 03 | Group-07 | Domain07

PowerGrid Reliability Intelligence: Early-Warning System for Grid Failure and Outage Prevention for PowerGrid Utilities

1. Business Problem Understanding and Objective Formulation

PowerGrid Utilities Corporation is a large transmission and distribution utility. Its network of substations, transformers, lines and smart-grid assets serves residential, commercial, industrial and government customers.



- **Business Problem:** Rising frequency of power outages, equipment failures, voltage instability events, and service interruptions affecting customers across multiple service territories resulting in significant financial losses, regulatory scrutiny, customer dissatisfaction, and reputational damage. Current monitoring systems are not acting upon time and are mostly reactive.
- **Business objective:** Transition to a predictive grid management strategy capable of reducing outages, improving grid reliability, identifying high-risk assets early, and optimizing maintenance spending before major service interruptions occur.
- **Analytical objectives:**
  - **Predictive Modelling:** Build predictive machine learning models to forecast outages. Train, evaluate and compare supervised machine-learning classifiers (Logistic Regression, Random Forest, XGBoost, AdaBoost) to predict grid_failure_flag (0 = normal operation vs 1 = failure).
  - **Risk Scoring & Segmentation:** Classify assets/substations into risk tiers (Low, Medium, High At-Risk) using probabilistic outputs.
  - **Feature Importance & Failure-Driver Analytics:** Identify the key drivers of grid failure (equipment health, inspection and oil condition, storm risk, maintenance gaps, voltage instability, load utilization) and their feature importance.
  - **Prescriptive Recommendation Engine:** Create automated maintenance and operational triggers (inspect, load-shift, pre-storm crew deployment, replace) based on model outputs and asset risk segments.

2. Challenges and Constraints

- Data quality issues include duplicate entries, missing sensor values, legacy system attributes, and class imbalance. Heterogeneous data collected across multiple departments introduces challenges in data integration and holistic analysis.
- Weather uncertainty and evolving grid demand patterns.
- Ensuring seamless real-time deployment and providing transparent, explainable predictions are key implementation constraints.
- Model explainability required for utility operators and regulators.

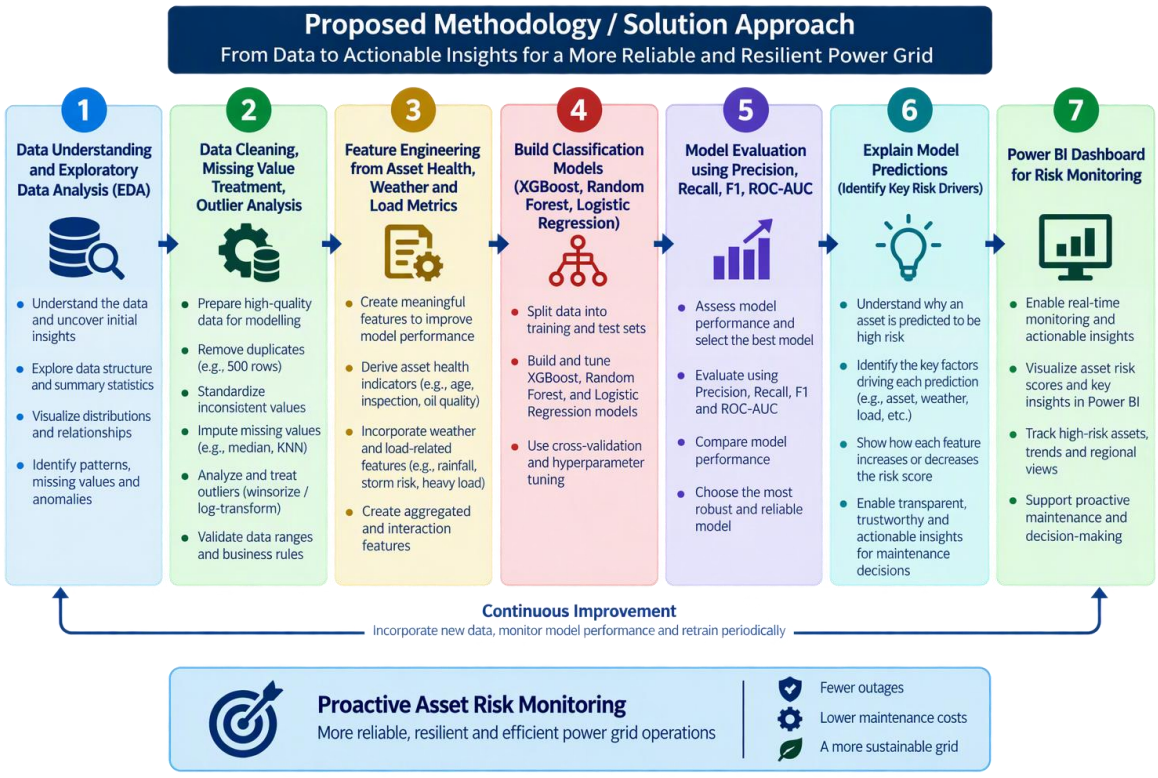
3. Background Study / Literature Review

Machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), and XGBoost have been widely applied for predictive maintenance, outage forecasting, asset health monitoring, and reliability assessment in power utility networks.

References:

- https://ieeexplore.ieee.org/abstract/document/6890363
- https://link.springer.com/article/10.1186/s42162-025-00574-w
- https://www.mdpi.com/2855728
- https://www.mdpi.com/1424-8220/24/13/4313

4. Proposed Methodology / Solution Approach



5. Proposed Timeline

Weeks	Activities	Milestone
1	Problem framing, literature review, pre-submission document	Pre-submission document
1–2	Data understanding and audit; cleaning and preprocessing	Clean dataset and data-quality log
2–3	EDA, hypothesis tests, feature engineering	EDA report; feature set v1
4–5	Baseline and advanced model development, tuning, cross-validation	Model comparison table
5–6	Evaluation, territory fairness audit, segmentation, risk tiers	Validated model and asset risk profiles
6–7	Dashboard, maintenance playbook, deployment proposal	Dashboard prototype
8	Final report, presentation, documentation and code hand-off	Final submission

6. Expected Deliverables / Outputs

Deliverable	Expected content
Data Analysis Report	Data quality, EDA, regional/asset/seasonal patterns, target definition, and limitations.
Machine Learning Model Package	Reproducible preprocessing pipeline, baseline models, selected model, evaluation, and calibration.
Dashboard / Visualizations	Reliability overview, asset risk distribution, failure of drivers, segments, and monitoring views.
Business Recommendations	Tiered maintenance and operations playbook, pilot design, operating thresholds, and reliability KPIs.
Final Presentation	Executive narrative covering problem, findings, model results, recommendations, and risks.
Technical Documentation	Data dictionary, feature definitions, code/run instructions, assumptions, metrics, and governance.
Deployment Proposal	Batch/API concept, SCADA and work-order integration points, engineer-review workflow, monitoring, and retraining criteria.